

WHAT We Are Doing

Simple understanding of the exact research question and scope

Research Project: Cross-Cycle Explanation Drift in Solar Flare Forecasting

Domain: Space Weather | Machine Learning | Explainable AI

Status: Locked research direction - October 2026

Purpose

Use this document to explain what the research actually studies, what is being compared, and what counts as a result.

Based on the locked Master Research Reference v1.0

1. Research in One Sentence

One-line answer

Hum Solar Cycle 24 ke SHARP active-region data par solar-flare prediction models train karenge, models ko freeze karenge, aur same models ko Cycle 24 holdout aur Cycle 25 par test karke prediction performance aur SHAP explanations dono compare karenge.

2. Research Problem

Normally ML flare forecasting mein hum poochte hain: **model prediction kitni sahi karta hai?**

Hum ek second question add kar rahe hain: **model prediction karte waqt kin magnetic features ko importance deta hai, aur kya woh pattern Solar Cycle 24 se Cycle 25 mein change hota hai?**

Core idea

Prediction stable reh sakti hai, lekin feature-attribution pattern drift kar sakta hai. Ya dono stable ho sakte hain. Ya dono degrade kar sakte hain. Hum result pehle assume nahi karte.

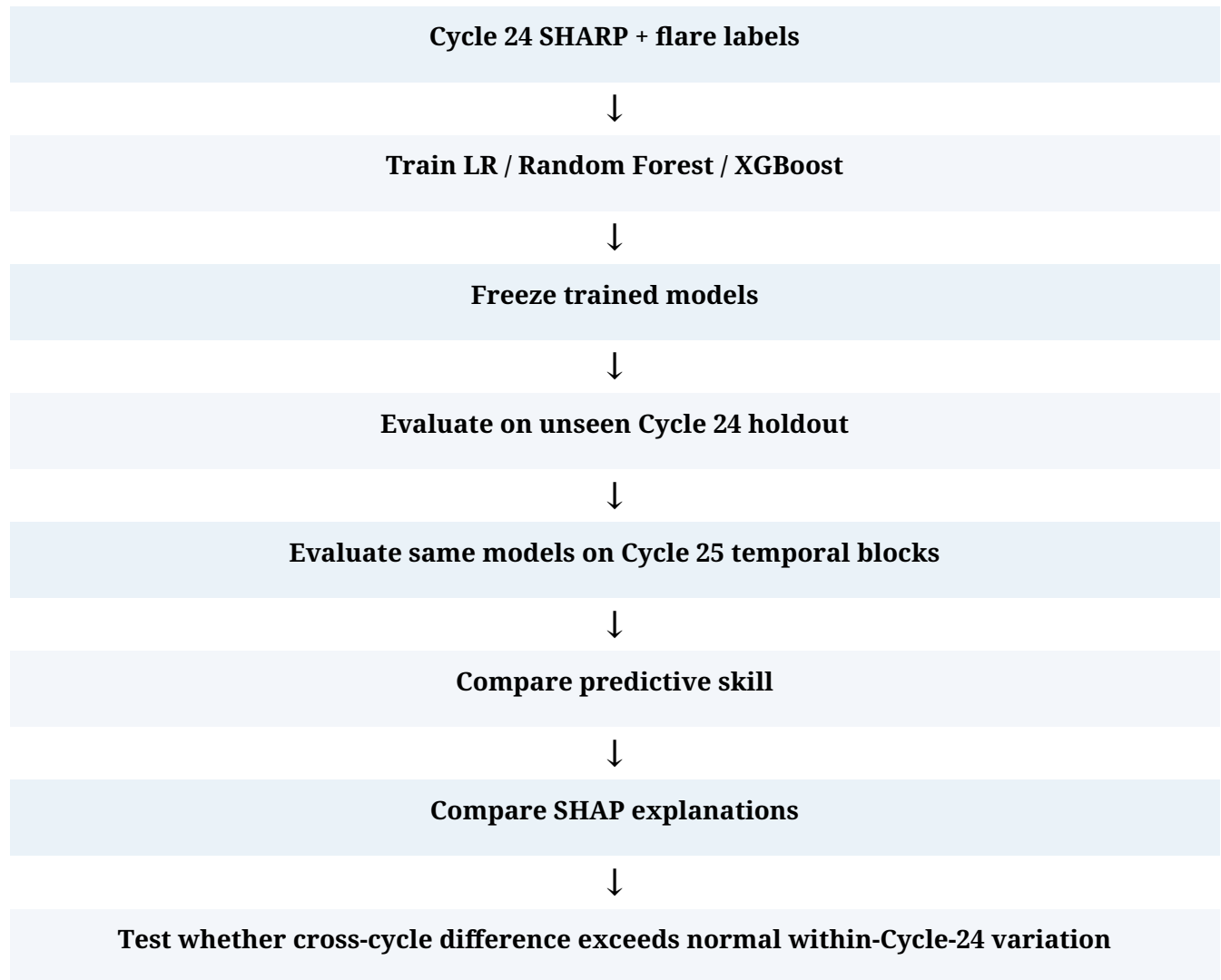
3. Important Terms - Very Simple

Concept	Simple meaning
Solar flare	Sun ke active region se sudden energetic event. Research target $\geq$ M-class flare hai.
Solar Cycle 24	Historical solar-activity regime jiska HMI/SHARP-era data training ke liye use hoga.
Solar Cycle 25	Naya solar-activity regime jahan frozen model ko out-of-distribution test karenge.
SHARP features	Active-region magnetic parameters, jaise R_VALUE, TOTUSJH aur USFLUX.
Frozen model	Training ke baad model ko Cycle 25 ke liye retrain ya fine-tune nahi karna.
SHAP	Model ke feature-attribution ko explain karne ka method: kis feature ne prediction mein kitna contribution diya.

Explanation drift

Same fixed model ka feature-attribution structure different evaluation regime mein statistically meaningful way mein change hona.

4. Exactly What We Compare



5. Two Main Outputs

A. Predictive stability - TSS plus complementary metrics tell us whether the model still forecasts well.

B. Explanation stability - SHAP statistics tell us whether the model is relying on a similar attribution structure across cycles.

6. Possible Final Outcomes

Observed pattern	Safe interpretation
Prediction stable + SHAP stable	Model transfers with similar aggregate skill and similar attribution structure.
Prediction stable + SHAP drifts	Most interesting explanation-drift case: similar skill, changed attribution structure.
Prediction drops + SHAP drifts	Both predictive generalization and explanation structure are unstable.
Prediction drops + SHAP stable	Model relies on similar features but loses skill; calibration/prevalence/feature-range shift may matter.

7. What We Are NOT Claiming

- We are not claiming this is the first cross-cycle solar-flare prediction study.
- We are not claiming this is the first use of SHAP for flare forecasting.
- We are not claiming SHAP proves solar physics or causality.
- We are not assuming explanation drift exists before running the experiment.
- We will not say "right for the wrong reasons" unless evidence supports a much stronger causal argument.

2-minute team version

Cycle 24 par model train -> freeze -> Cycle 24 holdout aur Cycle 25 par same model test -> TSS compare -> SHAP compare -> within-Cycle-24 null baseline se check kiya jayega ki SHAP change unusual hai ya normal variability. Final result stable ya drifting dono ho sakta hai.